

Prateek Nigam

Data Scientist

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SUMMARY

Data Scientist with over 5 years of experience in AI, multi-agent systems, and advanced language modeling, currently pursuing a Master's in Computer Science (AI) at the University of Galway. Proficient in Python, LLMs, RAG, and frameworks like LangChain and LangGraph, with a proven track record of optimizing AI-driven solutions for enterprise applications, including Market Mix Modeling. Adept at delivering scalable systems and collaborating cross-functionally to align data insights with business goals.

SKILLS

Technical Skills: Python, Transformers, Fine-Tuning, RAG, LLMs, Multi-Agent Systems (LangChain, LangGraph, Crew AI), Machine Learning (Regression, Decision Trees, LGBM), Deep Learning, Bayesian Regression, Cloud Platforms (AWS), Data Analysis, Data Visualization.

Strategic Problem Solving: Critical and analytical thinking, Collaborative problem-solving, aligning AI solutions with business objectives, rapid adaptation to new domains and technologies.

EXPERIENCE

Fractal Analytics

Oct 2021 - Aug 2024

Data Scientist

Mumbai, India

Market Mix Modelling

- Contributed as a **subject matter expert in MMM (Market Mix Modeling)** across CPG/FMCG product categories – analyzed marketing effectiveness – **optimized advertisement investments**, leading to a **projected 5-10 % increase in ROI** and profitability.
- Optimized **Hierarchical Bayesian Modeling** run time - **from 5 hours to 8 mins** by replacing theano tensors with Jax arrays - thus enhancing efficiency allowing to handling larger datasets and **increasing project capacity by 50%**.
- Faced the need for insightful and detailed **model at the zip code level** – researched and built technical models using **LGBM modeling** with a performance of around **70%** – helped improve targeted marketing.
- Addressed the **problem of redundant data analysis** – **automated the complete EDA process** – saved approximately **one-third in analysis time**, boosting overall productivity.

Copy Testing and Survey Analysis

- Conducted a copy testing exercise to analyze different datasets, focusing on **KPIs like customer retention, ad relevance, and brand appeal** – **identified effective source cross feature combinations** for predicting **Volume per GRP**.
- Built **regression models for kpi attribution** and created **decision trees** using information gain – **worked closely with the client** team to achieve an **F1 score of about 48%** through **manual tree creation and integration of business insights**.

Capgemini India Pvt Ltd.

Aug 2019 - Oct 2021

Software Engineer

Bangalore, India

- Collaborated** with cross-functional teams to design **solutions** based on evolving client requirements.
- Enhanced the PySpark transformation ETL tool, improving **data processing efficiency** and performance.
- Implemented **Single Sign-On (SSO)** using **Kerberos** and integrated LDAP for Active Directory group filtering.

EDUCATION

University Of Galway, Ireland

2024 – Current

Msc - Computer Science with Artificial Intelligence

Acropolis Institute of Technology and Research

2015 – 2019

B.Tech - Electronics and Communication Engineering

CGPA - 7.4/10

PROJECTS

- Building the Beast - Transformers from Scratch**  - Built transformer models from scratch to learn how they work. I broke down each part to understand how they fit together.
- AutoMMM**  - AI-powered Market Mix Modeling system using Python, LangChain, and multi-agent collaboration that autonomously analyzes ad spend, sales, and external factors using multi-agent collaboration with shared context.